

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Natural Sciences (N.Sc.)

MID-SEMESTER EXAMINATION
DURATION: 2 HOURS

WINTER SEMESTER, 2024-2025
FULL MARKS: 160

MATH 4141: Geometry and Differential Calculus

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 4 (four) questions. Marks of each question and corresponding CO and PO are written in the right margin with brackets.

1. (a) Consider the piecewise function $g(x)$ defined as follows:

[9]

$$g(x) = \begin{cases} x; & \text{if } x < 1 \\ 3; & \text{if } x = 1 \\ 2 - x^2; & \text{if } 1 < x \leq 2 \\ x - 3; & \text{if } x > 2 \end{cases}$$

(CO1)

(PO2)

Now, evaluate each of the following limits, if exists.

(i). $\lim_{x \rightarrow 1} g(x)$. ✓ exists

(ii). $\lim_{x \rightarrow 2^-} g(x)$. ↘

(iii). $\lim_{x \rightarrow 2^+} g(x)$. ↗

- (b) Define the continuity of a function. Let a function $f(x)$ be defined as follows:

[9]

$$f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}; & x < 2 \\ ax^2 - bx + 3; & 2 \leq x < 3 \\ 2x - a + b; & x \geq 3 \end{cases}$$

(CO1)

(PO1)

Find the values of the constants a and b that make the function $f(x)$ continuous everywhere.

- (c) (i). Find $f'(x)$ if $f(x) = (x^2 + 1)\left(x + 5 + \frac{1}{x}\right)$ using product rule.

[12]

(CO1)

- (ii). Find $g'(x)$ if $g(x) = \frac{4 - 3x}{3x^2 + x}$ using quotient rule.

(PO2)

- (iii). Find $\frac{dy}{dx}$ if $y = [1 + \sin^3(x^5)]^{12}$ using chain rule.

- (d) Compute the limits (if exist) using L'Hôpital's rule:

(i). $\lim_{x \rightarrow 0} \frac{\tan x - x}{x^3}$.

(ii). $\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$.

$\lim_{x \rightarrow 0} \left(\frac{\tan x}{x^3} - \frac{1}{x^2} \right) = \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \frac{1}{x^2} - \frac{1}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \frac{1}{x^2} - \frac{1}{x^2}$
 $= \lim_{x \rightarrow 0} \frac{\tan x}{x} \cdot \frac{1}{x^2} - \frac{1}{x^2}$

[10]

(CO1)

(PO2)

$\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$
 $= \lim_{x \rightarrow 1^+} \frac{(1 - \ln x)}{\ln x (x-1)}$
 $= \lim_{x \rightarrow 1^+} \frac{(1 - \ln x)}{\ln x (x-1)}$

2. (a) (i). If $y = [x + \sqrt{1+x^2}]^m$, show that $(x^2 + 1)y_{n+2} + (2n+1)xy_{n+1} + (n^2 - m^2)y_n = 0$ [20]
using Leibnitz's theorem. (CO1)
(ii). If $y = e^{ax} \sin(bx + c)$, then prove that (PO1)

$$y_n = e^{ax} (a^2 + b^2)^{\frac{n}{2}} \sin \left[bx + c + n \tan^{-1} \left(\frac{b}{a} \right) \right].$$

- (b) Use implicit differentiation technique to find $\frac{dy}{dx}$ for a Lemniscate represented [10]
by $2(x^2 + y^2)^2 = 25(x^2 - y^2)$. Hence, find the equation for the tangent line to the (CO2)
Lemniscate at the point (3,1). (PO2)

- (c) A camera is mounted at a point 3000 ft from the base of a rocket launching [10]
pad. If the rocket is rising vertically at 880 ft/s when it is 4000 ft above the (CO4)
launching pad, how fast must the camera elevation angle change at that instant (PO2)
to keep the camera aimed at the rocket?

3. (a) Consider the function $f(x) = x^4 - 4x^3 + 10$ [20]
(i). Find the intervals on which $f(x)$ is increasing. (CO1)
(ii). Find the intervals on which $f(x)$ is decreasing. (PO2)
(iii). Find the open intervals on which $f(x)$ is concave up.
(iv). Find the interval on which $f(x)$ is concave down.
(v). Find the critical points and inflection points, if any.

- (b) Find the relative extrema of the function $f(x) = -x^3 + 6x^2 + 36x - 60$. [10]
(CO2)
(PO2)

- (c) Find the absolute maximum and absolute minimum values of the function [10]
 $f(x) = 2x^3 - 3x^2 - 12x + 1$ on the interval $[-2, 3]$, and determine where these (CO2)
values occur. (PO2)

4. (a) Write down the statement of Rolle's Theorem. Verify Rolle's Theorem for the [15]
function $f(x) = \ln(4 + 2x - x^2)$ on the interval $[-1, 3]$. (CO1)
(PO2)

- (b) An open box is to be made from a 16-inch by 30-inch piece of card board by [15]
cutting out squares of equal size from the four corners and bending up the (CO2)
sides. What size should the squares be to obtain a box with the largest volume? (PO2)

- (c) (i). Find the second-order partial derivatives of $f(x, y) = 4x^2 - 8xy^4 + 7y^5 - 3$. [10]
(ii). Let $u(w, x, y, z) = xe^{yw} \sin^2 z$ be a given function, then find $\frac{\partial^4 u}{\partial x \partial y \partial w \partial z}$ and (CO2)
evaluate $\frac{\partial u}{\partial w}(0, 0, 1, \pi)$. (PO2)

$$\frac{\partial u}{\partial w}(0, 0, 1, \pi).$$

The End

$$\frac{\partial u}{\partial x} \times \frac{\partial u}{\partial y} \times \frac{\partial u}{\partial w} \times \frac{\partial u}{\partial z}$$